Lahore Garrison University

DHA Phase 6 Lahore

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Section	E
Subject	Cloud Computing
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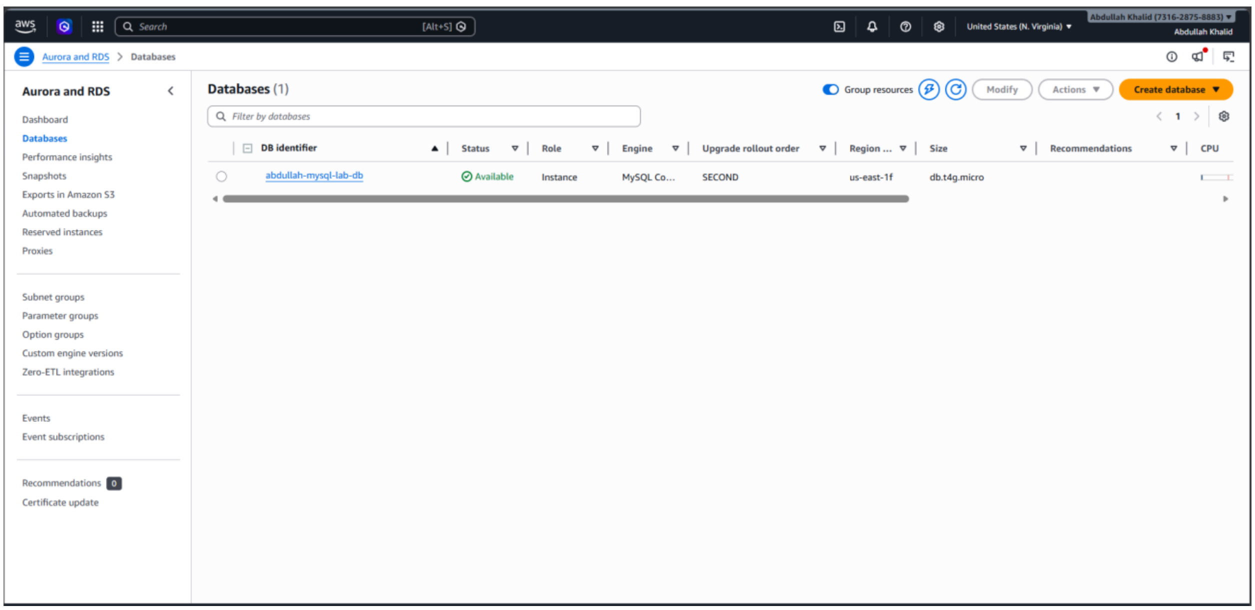
LAB 6

Database Management using Amazon RDS

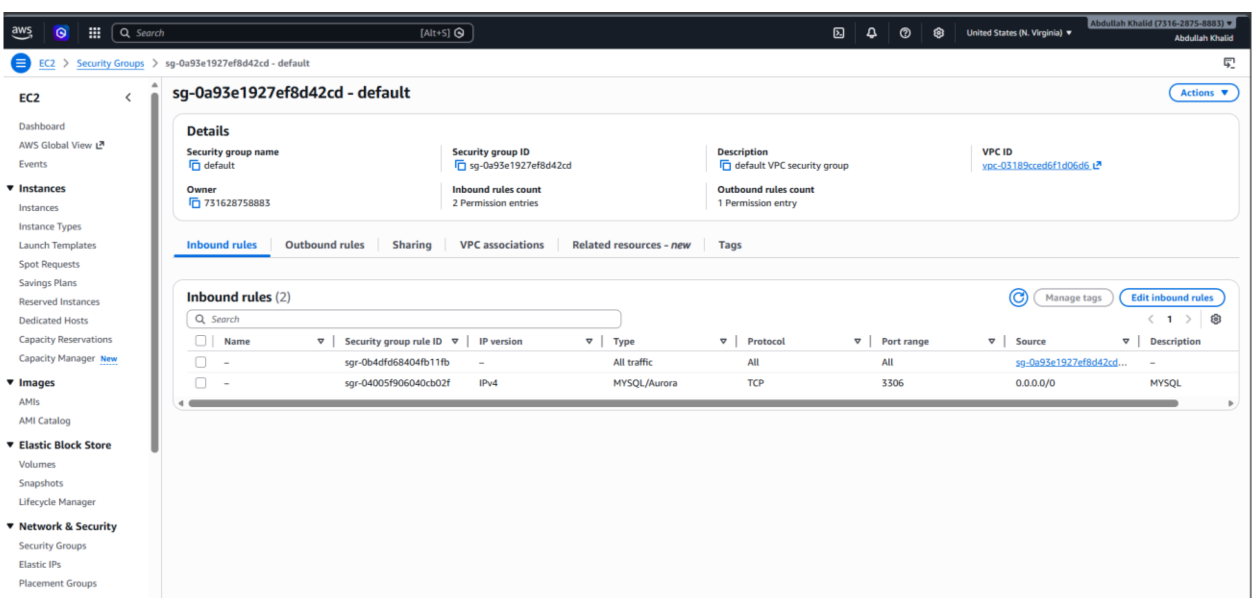
Question 1:

Create a database and insert multiple records.

Database Creation:



Security rule added to use Database in CloudShell:



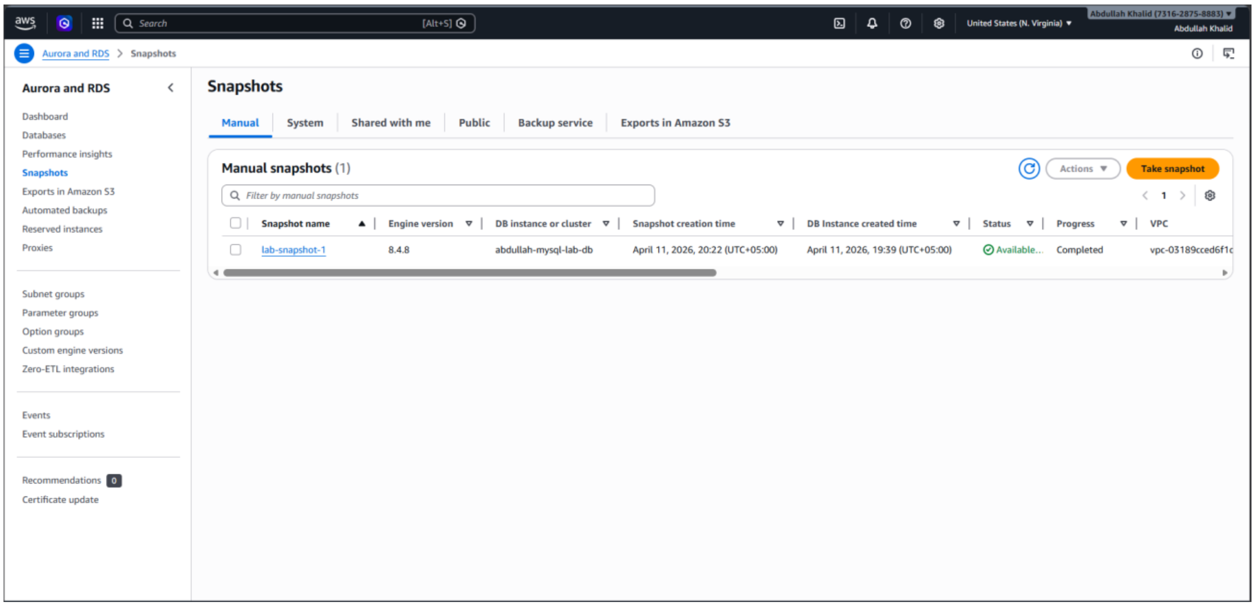
Inserting Multiple Records in Database:


```
~ $ mysql -h abdullah-mysql-lab-db.ck5waces0ayu.us-east-1.rds.amazonaws.com -P 3306 -u admin -pLabRDS123!
CREATE DATABASE labdb;
ERROR 2002 (HY000): Can't connect to MySQL server on 'abdullah-mysql-lab-db.ck5waces0ayu.us-east-1.rds.amazonaws.com' (115)
~ $ CREATE DATABASE labdb;
-bash: CREATE: command not found
~ $ mysql -h abdullah-mysql-lab-db.ck5waces0ayu.us-east-1.rds.amazonaws.com -P 3306 -u admin -pLabRDS123!


^[[A^[[ERROR 2002 (HY000): Can't connect to MySQL server on 'abdullah-mysql-lab-db.ck5waces0ayu.us-east-1.rds.amazonaws.com' (115)
~ $
~ $ mysql -h abdullah-mysql-lab-db.ck5waces0ayu.us-east-1.rds.amazonaws.com -P 3306 -u admin -pLabRDS123!
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 42
Server version: 8.4.8 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> CREATE DATABASE labdb;
Query OK, 1 row affected (0.037 sec)

MySQL [(none)]> USE labdb;
Database changed
MySQL [labdb]> CREATE TABLE students(id INT PRIMARY KEY AUTO_INCREMENT, name VARCHAR(50));
Query OK, 0 rows affected (0.049 sec)

MySQL [labdb]> INSERT INTO students(name) VALUES('Ali'),('Ahmed'),('Sara'),('Fatima');
Query OK, 4 rows affected (0.026 sec)
Records: 4  Duplicates: 0  Warnings: 0

MySQL [labdb]> SELECT * FROM students;
+----+--------+
| id | name   |
+----+--------+
|  1 | Ali    |
|  2 | Ahmed  |
|  3 | Sara   |
|  4 | Fatima |
+----+--------+
4 rows in set (0.002 sec)

MySQL [labdb]>
```

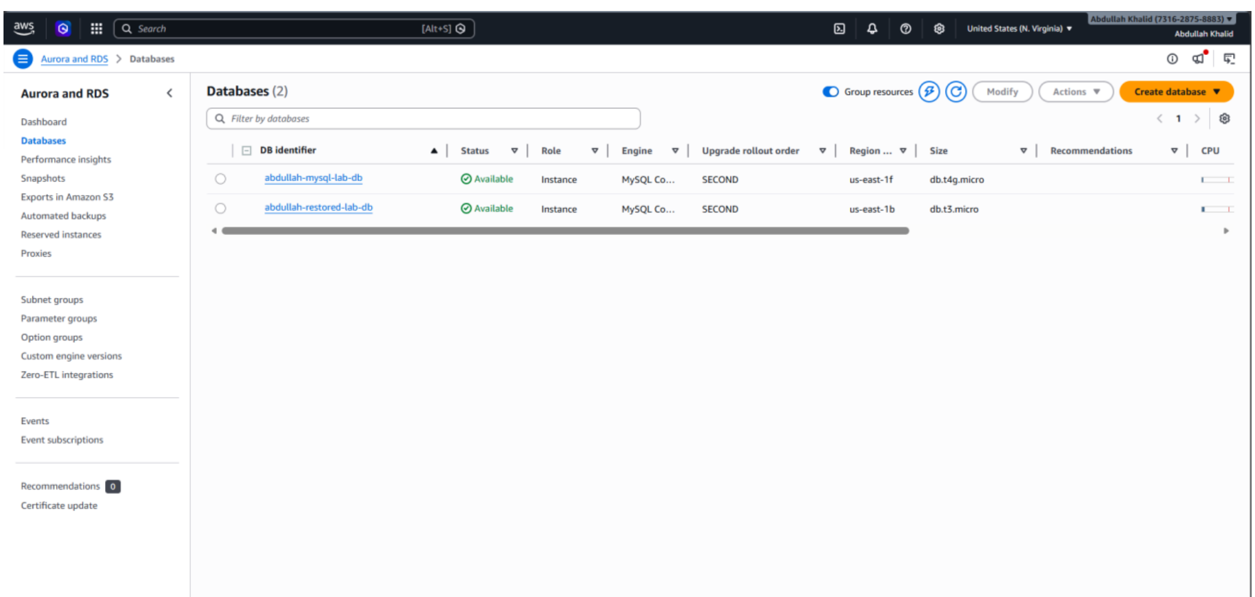
Question 2:

Take a snapshot and restore database:

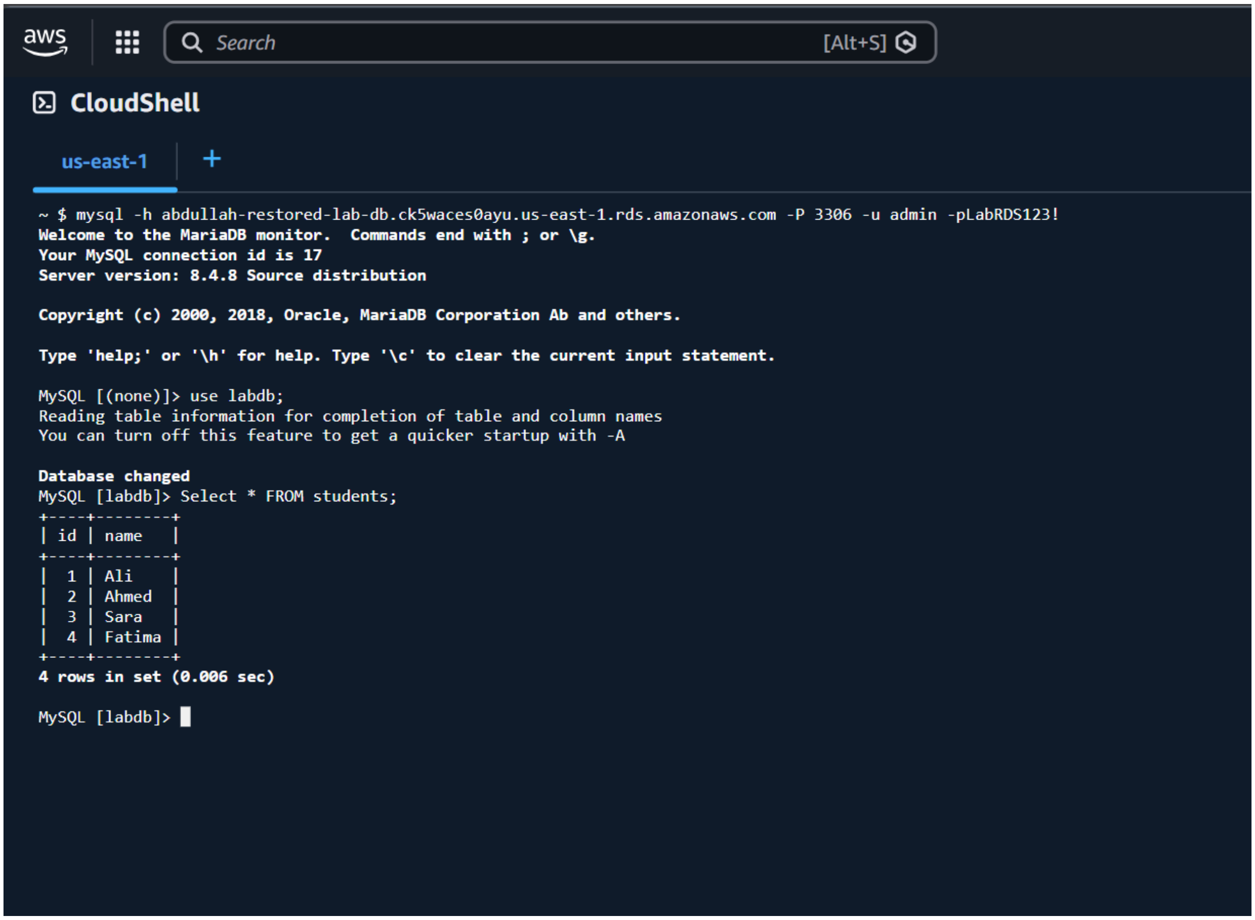
Taking Snapshot:



Restoring Database:



Same Data showing in this restored Database:



Question 3:

Enable Multi-AZ and explain its benefit.

Benefits:

Production uses **db.t3.micro+** for:

- Automatic failover <60s if AZ fails
- Zero data loss (synchronous replication)
- No downtime during backups/patching

Multi-AZ unavailable on db.t4g.micro free tier: